

## Mark schemes

### Q1.

|                      | Marking Instructions                                                                                                             | AO     | Marks | Typical Solution                                                                                                                                           |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------|--------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a)                  | Selects appropriate method for example by changing to reduced equation by dividing by $\tan x$                                   | AO3.1a | M1    | $\frac{dy}{dx} + (\cot x)y = \sin x$                                                                                                                       |
|                      | Finds correct integrating factor                                                                                                 | AO1.1b | B1    | Integrating factor $= e^{\int (\cot x) dx}$<br>$= e^{\ln(\sin x)} = \sin x$                                                                                |
| ALT                  | Alt. finds an integrating factor by inspection, using original equation. (PI)                                                    | AO3.1a | M1    | $\cos x \tan x \frac{dy}{dx} + y \cos x = \sin x \tan x \cos x$<br>$\sin x \frac{dy}{dx} + (\cos x)y = \sin^2 x$                                           |
|                      | Finds integrating factor = $\cos x$                                                                                              | AO1.1b | B1    |                                                                                                                                                            |
|                      | Multiplies reduced or original equation by 'their' integrating factor and identifies LHS as differential of $y \times \sin x$ PI | AO1.1a | M1    | $\sin x \frac{dy}{dx} + (\cos x)y = \sin^2 x$<br>$\frac{d}{dx} [y \sin x] = \sin^2 x$                                                                      |
|                      | Uses appropriate integration method for RHS of 'their' equation                                                                  | AO1.1a | M1    | $y \sin x = \int \left\{ \frac{1}{2} (1 - \cos 2x) \right\} dx$                                                                                            |
|                      | Integrates correctly to obtain correct solution                                                                                  | AO1.1b | A1    | $y \sin x = \frac{1}{2} x - \frac{1}{4} \sin 2x + C$                                                                                                       |
|                      | Uses boundary condition after integration completed in either $y \sin x = \dots$ or $y = \dots$ form OE                          | AO1.1a | M1    | $\sin \frac{\pi}{4} \cdot \frac{1}{2\sqrt{2}} = \frac{1}{2} \cdot \frac{\pi}{4} - \frac{1}{4} \sin \frac{\pi}{2} + C$<br>$C = \frac{1}{2} - \frac{\pi}{8}$ |
| (b)                  | States fully correct particular solution                                                                                         | AO1.1b | A1    | $y \sin x = \frac{1}{2} x - \frac{1}{4} \sin 2x + \frac{1}{2} - \frac{\pi}{8}$                                                                             |
| <b>Total 7 marks</b> |                                                                                                                                  |        |       |                                                                                                                                                            |

### Q2.

(a)  $\frac{dy}{dx} + \frac{\sec^2 x}{\tan x} y = \tan x$

IF is  $\exp\left(\int \frac{\sec^2 x}{\tan x} dx\right)$   
*and with integration attempted*

M1

$= e^{\ln(\tan x)} = \tan x$   
*AG Be convinced*

A1

2

(b)  $\tan x \frac{dy}{dx} + (\sec^2 x)y = \tan^2 x$

$\frac{d}{dx} [y \tan x] = \tan^2 x$   
*LHS as differential of  $y \times IF$  PI*

M1

$y \tan x = \int \tan^2 x dx$

A1

$\Rightarrow y \tan x = \int (\sec^2 x - 1) dx$   
*Using  $\tan^2 x = \sec^2 x - 1$  PI or other valid methods to integrate  $\tan^2 x$*

m1

$y \tan x = \tan x - x (+c)$   
*Correct integration of  $\tan^2 x$ ; condone absence of  $+c$ .*

A1

$3 \tan \frac{\pi}{4} = \tan \frac{\pi}{4} - \frac{\pi}{4} + c$

*Boundary condition used in attempt to find value of  $c$*

m1

$c = 2 + \frac{\pi}{4}$  so  $y \tan x = \tan x - x + 2 + \frac{\pi}{4}$

$y = 1 + (2 - x + \frac{\pi}{4}) \cot x$   
*ACF*

A1

Q3.

(a)  $\frac{d}{dx} [\ln(\ln x)] = \frac{1}{\ln x} \times \frac{1}{x}$   
ACF

B1

1

(b) (i)  $\frac{dy}{dx} + \frac{1}{x \ln x} y = 9x^2$

An IF is  $\exp \left\{ \int [1 / (x \ln x)] (dx) \right\}$   
... and with integration attempted

M1

$= e^{\ln(\ln x)} = \ln x$

AG Must see  $e^{\ln(\ln x)}$  before  $\ln x$

A1

2

(ii)  $\ln x \frac{dy}{dx} + \frac{1}{x} y = 9x^2 \ln x$

$\frac{d}{dx} [y \ln x] = 9x^2 \ln x$

LHS as differential of  $y \times \ln x$  PI

M1

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$y \ln x = \int 9x^2 \ln x \, dx$

A1

$\Rightarrow y \ln x = \int \ln x \, d[3x^3] = 3x^3 \ln x - \int 3x^3 \left( \frac{1}{x} \right) dx$

$\int kx^2 \ln x \, (dx) = px^3 \ln x - \int px^3 \left( \frac{1}{x} \right) (dx) \text{ or better}$

m1

$y \ln x = 3x^3 \ln x - x^3 (+c)$

ACF Condone missing '+c'

A1

When  $x = e$ ,  $y = 4e^3$ ,  $4e^3 = 3e^3 - e^3 + c$

*Dep on previous M1m1. Boundary condition used in attempt to find value of 'c' after integration is completed*

m1

$$c = 2e^3$$

$$\Rightarrow y \ln x = 3x^3 \ln x - x^3 + 2e^3$$

$$y = 3x^3 - \frac{(x^3 - 2e^3)}{\ln x}$$

ACF

A1

6

[9]

**Q4.**

(a) IF is  $\exp\left(\int \frac{2}{x} dx\right)$   
and with integration attempted

M1

$$= e^{2\ln x}$$

PI

A1

$$= x^2$$

A1

$$\frac{d}{dx} [yx^2] = x^2 \ln x$$

LHS; PI

M1

$$\Rightarrow yx^2 = \int (\ln x) \frac{d}{dx} \left( \frac{x^3}{3} \right)$$

*Attempt integration by parts in correct direction to integrate  $x^p \ln x$*

M1

$$= \frac{x^3}{3} \ln x - \int \frac{x^3}{3} dx$$

RHS

A1

$$yx^2 = \frac{x^3}{3} \ln x - \frac{x^3}{9} + A$$

$$\{y = \frac{x}{3} \ln x - \frac{x}{9} + Ax^{-2}\}$$

A1

3

(b) Now, as  $x \rightarrow 0$ ,  $x^k \ln x \rightarrow 0$

*Must be stated explicitly for a value of  $k > 0$*

E1

As  $x \rightarrow 0$ ,  $y \rightarrow 0 \Rightarrow A = 0$

*Const of int = 0 must be convincing*

B1

$$yx^2 = \frac{x^3}{3} \ln x - \frac{x^3}{9}$$

$$\text{When } x = 1, y = -\frac{1}{9}$$

*ft on one slip but must have made a realistic attempt to find A*

B1F

3

[10]

Q5.

(a) IF is  $e^{\int \frac{4}{2x+1} dx}$   
PI

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M1

$$e^{2 \ln(2x+1) (+c)} = e^{\ln(2x+1)^2 (+c)}$$

*Either O.E. Condone missing '+c'*

A1

$$= (A)(2x+1)^2$$

*Ft on earlier  $e^{\ln(2x+1)}$ , condone missing 'A'*

A1F

$$(2x+1)^2 \frac{dy}{dx} + 4(2x+1)y = 4(2x+1)^7$$

$$\frac{d}{dx}[(2x+1)^2 y] = 4(2x+1)^7$$

LHS as  $d/dx (y \times c's IF) PI$  **and** also RHS of form  $p(2x+1)^q$

M1

$$(2x+1)^2 y = \int 4(2x+1)^7 dx$$

A1

$$(2x+1)^2 y = \frac{1}{4} (2x+1)^8 + c$$

Correct integration of  $p(2x+1)^q$  to

$$\frac{p(2x+1)^{q+1}}{2(q+1)} \quad (+c) \text{ ft for } q > -2 \text{ only}$$

B1F

$$(GS): y = \frac{1}{4} (2x+1)^6 + c(2x+1)^{-2}$$

Must be in the form  $y = f(x)$ , where  $f(x)$  is ACF

A1

$$(b) \quad y = \frac{1}{4} (2x+1)^6 + c(2x+1)^{-2}$$

$$\text{When } x=0, \frac{dy}{dx} = 0$$

Using boundary condition  $x=0, \frac{dy}{dx} = 0$

and c's GS in (a) towards obtaining a value for c

M1

$$\Rightarrow y = 1 \left[ \frac{dy}{dx} 3(2x+1)^5 - 4c(2x+1)^{-3} \right]$$

Either  $y = 1$  or correct expression for  $dy/dx$  in terms of  $x$  only

B1

$$\Rightarrow c = \frac{3}{4} \text{ so } y = \frac{1}{4} (2x+1)^6 + \frac{3}{4} (2x+1)^{-2}$$

CSO

A1

3

[10]

**Q6.**

$$IF \text{ is } e^{\int -\frac{2}{x} dx}$$

M1

$$= e^{-2\ln(x)} (+c) = e^{\ln(x)^{-2}} (+c)$$

OE Condone missing c

A1

$$= (k)x^{-2}$$

Ft earlier sign error

A1F

$$x^{-2} \frac{dy}{dx} - 2x^{-3}y = 2xe^{2x}$$

$$\frac{d}{dx}(x^{-2}y) = 2xe^{2x}$$

LHS as d/dx(y × IF) PI

M1

$$x^{-2}y = \int 2xe^{2x} dx$$

$$= \int x d(e^{2x}) = x e^{2x} - \int e^{2x} dx$$

Integration by parts in correct dirn

M1

A1

$$x^2 y = x e^{2x} - \frac{1}{2} e^{2x} (-c)$$

ACF

A1

When  $x = 2$ ,  $y = e^4$  so

$$\frac{1}{4}e^4 = 2e^4 - \frac{1}{2}e^4 + c$$

Boundary condition used to find c after integration.

m1

$$c = -\frac{5}{4}e^4$$

$$y = x^3 e^{2x} - \frac{1}{2} x^2 e^{2x} - \frac{5}{4} x^2 e^4$$

*Must be in the form  $y = f(x)$*

A1

[9]

**Q7.**

$$\frac{dy}{dx} + (\cot x)y = \sin 2x$$

$$\text{IF is exp } \left( \int \cot x dx \right)$$

*and with integration attempted*

M1

$$= e^{\ln(\sin x) (+c)}$$

*OE Condone missing '+c'*

A1

$$= (k) \sin x$$

*'IF = sin x' scores M1A1A1*

A1

$$\sin \frac{dy}{dx} + (\cos x)y = \sin 2x \sin x$$

$$\frac{d}{dx} [y \sin x] = \sin 2x \sin x$$

*LHS as differential of  $y \times \text{IF}$  PI*

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$$y \sin x = \int \sin 2x \sin x dx$$

*Ft on c's IF provided no exp. or logs*

A1F

$$\Rightarrow y \sin x = \int 2 \sin^2 x \cos x dx$$

*$\sin 2x = 2 \sin x \cos x$  used*

B1

$$\Rightarrow y \sin x = \int 2 \sin^2 x d(\sin x)$$

*dep on both Ms*

*Use of relevant substitution to stage  $\int 2s^2 ds$   
or further or by inspection to  $k \sin^3 x$*



m1

$$y \sin x = \frac{2}{3} \sin^3 x (+c)$$

ACF

A1

$$\frac{1}{2} \sin \frac{\pi}{6} = \frac{2}{3} \sin^3 \frac{\pi}{6} + c$$

dep on both Ms

Boundary condition used in attempt to find value of c after integration

m1

$$c = \frac{1}{6} \text{ so } y \sin x = \frac{2}{3} \sin^3 x + \frac{1}{6}$$

CSO – no errors seen – accept equivalent forms

A1

[10]

Q8.

IF is exp  $\left(\int \frac{3}{x} dx\right)$

and with integration attempted

M1

$$= e^{3 \ln x}$$

PI

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A1

$$= x^3$$

A1

$$\frac{d}{dx} [yx^3] = x^3 (x^4 + 3)^{\frac{3}{2}}$$

LHS. Use of c's IF. PI

M1A1

$$\Rightarrow yx^3 = \frac{1}{10} (x^4 + 3)^{\frac{5}{2}} + A$$

$$k(x^4 + 3)^{\frac{5}{2}}$$

m1

Condone missing 'A'

A1

$$\Rightarrow \frac{1}{5} = \frac{1}{10} (4)^{\frac{5}{2}} + A$$

Use of boundary conditions in attempt to find constant after intgr. Dep on two M marks, not dep on m

m1

$$\Rightarrow A = -3; \quad (*)$$

$$\Rightarrow yx^3 = \frac{1}{10} (x^4 + 3)^{\frac{5}{2}} - 3$$

ACF. The A1 can be awarded at line (\*) provided a correct earlier eqn in y, x and 'A' is seen immediately before boundary conditions are substituted.

A1

[9]

Q9.

(a) IF is  $e^{\int -\frac{2}{x} dx}$   
 $e^{\int \pm \frac{2}{x} dx}$

M1

$$= e^{-2 \ln x}$$

PI

A1

$$= e^{\ln x^{-2}} = x^{-2} = \frac{1}{x^2}$$

AG Be convinced

A1

3

(b)  $\frac{d}{dx} \left( \frac{y}{x^2} \right) = \frac{1}{x^2} x$

LHS as d/dx (y × IF)

M1

PI

A1

$$\frac{y}{x^2} = \int \frac{1}{x} dx = \ln x + c$$

RHS Condone missing '+ c' here

A1

$$y = x^2 \ln x + cx^2$$

A1

4

[7]

Q10.

IF is  $e^{\int -\tan x dx}$

Award even if negative sign missing

$$= e^{\ln(\cos x) (+c)}$$

OE Condone missing c

M1

$$= (k) \cos x$$

ft earlier sign error

A1

A1F

$$\cos x \frac{dy}{dx} - y \tan x \cos x = 2 \sin x \cos x$$

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$$\frac{d}{dx}(y \cos x) = 2 \sin x \cos x$$

LHS as  $\frac{d}{dx}(y \times IF)$  PI

M1

$$y \cos x = \int 2 \sin x \cos x dx$$

ft on c's IF provided no exp or logs

A1F

$$y \cos x = \int \sin 2x dx$$

Double angle or substitution OE for integrating  $2 \sin x \cos x$

m1

$$y \cos x = -\frac{1}{2} \cos 2x + c$$

ACF

A1

$$2 = -\frac{1}{2} + c$$

Boundary condition used to find c

m1

$$c = \frac{5}{2}$$

$$y \cos x = -\frac{1}{2} \cos 2x + \frac{5}{2}$$

ACF eg  $y \cos x = 2 + \sin^2 x$   
Apply ISW after ACF

A1

[9]

Q11.

IF is  $e^{\int \frac{4x}{x^2+1} dx}$

M1

$$= e^{2 \ln(x^2+1)}$$

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A1

$$= e^{\ln(x^2+1)^2} = (x^2 + 1)^2$$

Then  $e^{2 \ln(x^2+1)}$

A1ft

$$\frac{d}{dx} (y(x^2 + 1)^2) = x(x^2 + 1)^2$$

LHS as  $d/dx (y \times \text{cand's IF})$  PI and also  
RHS of form  $kx(x^2 + 1)^p$

M1A1ft

$$y(x^2 + 1)^2 = \int x(x^2 + 1)^2 dx$$

$$y(x^2 + 1)^2 = \frac{1}{6} (x^2 + 1)^3 + c$$



Use of suitable substitution to find RHS

or reaching  $k(x^2 + 1)^3$  OE

Condone missing c

M1A1

$$y(0) = 1 \Rightarrow c = \frac{5}{6}$$

m1

$$y = \frac{1}{6}(x^2 + 1) + \frac{5}{6(x^2 + 1)^2}$$

Accept other forms of  $f(x)$ 

$$\text{eg } y = \frac{\left(\frac{x^6}{6} + \frac{2x^4}{4} + \frac{x^2}{2} + 1\right)}{(x^2 + 1)^2}$$

A1

[9]

Q12.

$$(a) \quad u = \frac{dy}{dx} \Rightarrow \frac{du}{dx} = \frac{d^2y}{dx^2}$$

M1

$$x \frac{du}{dx} - u = 3x^2 \Rightarrow \frac{du}{dx} - \frac{1}{x}u = 3x$$

AG Substitution into LHS of DE and completion

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A1

2

$$(b) \quad \text{IF is exp} \left( \int -\frac{1}{x} dx \right)$$

and with integration attempted

M1

$$= e^{-\ln x}$$

A1

$$= x^{-1} \text{ or } \frac{1}{x}$$

or multiple of  $x^{-1}$ 

A1

$$\frac{d}{dx} [ux^{-1}] = 3$$

*LHS as differential of  $u \times \text{IF}$ . PI*

M1

$$\Rightarrow ux^{-1} = 3x + A$$

*Must have an arbitrary constant  
(Dep. on previous M1 only)*

m1

$$u = 3x^2 + Ax$$

A1

6

(c)  $\frac{dy}{dx} = 3x^2 + Ax$

*Replaces  $u$  by  $\frac{dy}{dx}$  and attempts to integrate*

M1

$$y = x^3 + \frac{Ax^2}{2} + B$$

*ft on cand's  $u$  but solution must have two arbitrary constants*

A1F

2

EXAM PAPERS PRACTICE [10]

**Q13.** ©2025 Exam Papers Practice. All rights reserved.

(a) IF is  $\exp \left( \int \frac{2}{x} dx \right)$

*And with integration attempted*

M1

$$= e^{2\ln x}$$

A1

$$= x^2$$

*CSO **AG** be convinced*

A1

3

(b)  $\frac{d}{dx} [yx^2] = 3x^2 (x^3 + 1)^{\frac{1}{2}}$

PI

M1A1

$$\Rightarrow yx^2 = \frac{2}{3}(x^3 + 1)^{\frac{3}{2}} + A$$

$$k(x^3 + 1)^{\frac{3}{2}}$$

m1

Condone missing 'A'

A1

$$\Rightarrow 4 = \frac{2}{3}(9)^{\frac{3}{2}} + A$$

Use of boundary conditions to find constant

m1

$$\Rightarrow A = -14$$

$$\Rightarrow y = x^{-2} \left\{ \frac{2}{3}(x^3 + 1)^{\frac{3}{2}} - 14 \right\}$$

Any correct form

A1

6

[9]

Q14.

IF is  $e^{\int \tan x dx}$

M1

$$e^{-\ln \cos x} = e^{\ln \sec x}$$

Accept either

A1

$$= \sec x$$

ft on earlier sign error

A1ft

$$\frac{d}{dx}(y \sec x) = \sec^2 x$$

M1A1

$$y \sec x = \int \sec^2 x dx$$

$$y \sec x = \tan x + c$$

*Condone missing c*

A1

$$y = 3 \text{ when } x = 0 \Rightarrow 3 \sec 0 = 0 + c$$

m1

$$c = 3 \Rightarrow y \sec x = \tan x + 3$$

*OE; condone solution finishing at  $c = 3$   
provided no errors*

A1

[8]

**Q15.**

(a) (i)  $y(1.1) = y(1) + 0.1[1 \ln 1 + 1/1]$

M1A1

$$= 1 + 0.1 = 1.1$$

A1

3

(ii)  $y(1.2) = y(1) + 2(0.1)[f(1.1, y(1.1))]$

M1A1

$$= 1 + 2(0.1)[1.1 \ln 1.1 + (1.1/1.1)]$$

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*On answer to (a)(i)*

A1 ✓

$$= 1 + 0.2 \times 1.104841198 \dots$$

$$= 1.22096824 = 1.221 \text{ to 3 dp}$$

CAO

A1

4

(b) (i) IF is  $e^{\int -\frac{1}{x} dx}$

*Condone  $e^{\int \frac{1}{x} dx}$  for M mark*

M1

$$= e^{-\ln x}$$



A1

$$= e^{\ln x - 1} = x^{-1} = \frac{1}{x}$$

**AG** (be convinced)

(b)(i) Solutions using the printed answer must be convincing before any marks are awarded

A1

3

(ii)  $\frac{d}{dx} \left( \frac{y}{x} \right) = \ln x$

M1A1

$$\frac{y}{x} = \int \ln x \, dx = x \ln x - \int x \left( \frac{1}{x} \right) dx$$

Integration by parts for  $x^k \ln x$

M1

$$\frac{y}{x} = x \ln x - x + c$$

Condone missing c.

A1

$$y(1) = 1 \Rightarrow 1 = \ln 1 - 1 + c$$

Dependent on at least one of the two previous M marks

m1

$$\Rightarrow c = 2 \Rightarrow y = x^2 \ln x - x^2 + 2x$$

$$\text{OE eg } \frac{y}{x} = x \ln x - x + 2$$

A1

6

(iii)  $y(1.2) = 1.222543... = 1.223$  to 3dp

B1

1

[17]

**Q16.**

(a) IF is  $e^{\int c \, dx}$

M1

$$= e^{\ln \sin x}$$

A1

$$= \sin x$$

AG

A1

3

(b)  $\frac{d}{dx}(y \sin x) = 2 \sin x \cos x$

M1 A1

$$y \sin x = \int \sin 2x \, dx$$

Method to integrate  $2 \sin x \cos x$

M1

$$y \sin x = -\frac{1}{2} \cos 2x + c$$

OE

A1

$$y = 2 \text{ when } x = \frac{\pi}{2} \Rightarrow 2 \sin \frac{\pi}{2} = -\frac{1}{2} \cos \pi + c$$

Depending on at least one M

m1

$$c = \frac{3}{2} \Rightarrow y \sin x = \frac{1}{2}(3 - \cos 2x)$$

OE eg  $y \sin x = \sin^2 x + 1$

A1

6

[9]

**Q17.**

(a)  $u = \frac{dy}{dx} + 2y \Rightarrow \frac{du}{dx} = \frac{d^2 y}{dx^2} + 2 \frac{dy}{dx}$

2 terms correct

M1

A1

$$\text{LHS of DE} \Rightarrow \frac{du}{dx} - 2 \frac{dy}{dx} + 4 \frac{dy}{dx} + 4y$$

$$\text{LHS: } \frac{du}{dx} + 2(u - 2y) + 4y$$

*Substitution into LHS of DE as far as no derivatives of y*

M1

$$\Rightarrow \frac{du}{dx} + 2u = e^{-2x}$$

CSO AG

A1

4

(b) IF is  $e^{\int 2dx} = e^{2x}$

B1

$$\frac{d}{dx}[ue^{2x}] = 1$$

M1 A1

$$\Rightarrow ue^{2x} = x + A$$

A1

$$\Rightarrow u = xe^{-2x} + Ae^{-2x}$$

A1

5

**Alternative : Those using CF+PI**

Auxiliary equation,  $m + 2 = 0 \Rightarrow u_{CF} = Ae^{-2x}$

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B1

For  $u_{PI}$  try  $u_{PI} = kxe^{-2x} \Rightarrow$

M1

$$ke^{-2x} - 2kxe^{-2x} + 2kxe^{-2x} (= e^{-2x})$$

LHS

A1

$$\Rightarrow k = 1 \Rightarrow u_{PI} = xe^{-2x}$$

A1

$$\Rightarrow u_{GS} = Ae^{-2x} + xe^{-2x}$$

A1

$$(c) \Rightarrow \frac{dy}{dx} + 2y = xe^{-2x} + Ae^{-2x}$$

Use (b) to reach a 1<sup>st</sup> order DE in y and x

M1

$$\text{IF is } e^{\int 2dx} = e^{2x}$$

B1

$$\Rightarrow \frac{d}{dx} [ye^{2x}] = x + A$$

A1ft

$$\Rightarrow ye^{2x} = \frac{x^2}{2} + Ax + B$$

A1ft

$$\Rightarrow y = e^{-2x} \left( \frac{x^2}{2} + Ax + B \right)$$

A1

5

[14]

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## Examiner reports

### Q2.

Most of the students were able to show that  $\tan x$  was an integrating factor, although a significant minority failed to realise that  $\int \frac{\sec^2 x}{\tan x} dx$  gave  $\ln \tan x$  directly and instead wrote the integrand in the form  $\cot x + \tan x$  and after integrating correctly applied the relevant log law and eventually, after further manipulation, showed that  $\tan x$  was an integrating factor. The small number of students who wrote the integrand as  $2\operatorname{cosec} 2x$  were generally less convincing in reaching the printed result. Almost all students showed that they knew how to use the printed integrating factor to start to solve the first order differential equation but a minority of these students could not then find a correct method to integrate  $\tan^2 x$ .

### Q3.

This question, which tested use of an integrating factor to solve a first-order differential equation, was a good source of marks for many students. The first two parts of the question were usually answered correctly and almost all students correctly reached the stage  $y \ln x = \int 9x^2 \ln x dx$  in part (b)(ii). A large majority of students used integration by parts correctly but a small number forgot the '9' at this stage and so lost the two remaining accuracy marks. Very few failed to include the '+c' and almost all of those students who had scored the method mark for their integration also scored the method mark for correctly using the given boundary condition. Surprisingly there was a minority of students who failed to score the final mark due to a miscopy of an earlier correct expression when substituting their value for  $c$ , as illustrated by  $y \ln x = 3x^2 \ln x - x^3 + 2e^3$ .

### Q4.

Most candidates were able to show that they knew how to find and use an integrating factor to solve a first order differential equation. The method of integration by parts was generally both spotted and applied correctly, although some failed to include the constant of integration in their general solution, thus ending with a general solution of a first order differential equation which contained no arbitrary constant.

In part (b) the examiners expected candidates to pay particular attention to the limit of  $x \ln x$  (or  $x^3 \ln x$ ) as  $x$  tended to 0. Also some candidates were less than convincing in dealing with the term  $C / x^2$  as  $x$  tended to 0. In general, part (b) was not answered as well as part (a).

### Q5.

It was pleasing to see that once again most students were able to show that they knew how to find and use a correct integrating factor to solve a first order differential equation. The more common errors were failure to multiply the right-hand side by the integrating factor before integrating, forgetting to divide the constant of integration by the integrating factor when rearranging the general solution into the form  $y = f(x)$  or even forgetting to include a constant of integration. Students who answered part (a) correctly were generally successful in part (b).

**Q6.**

Most candidates were able to show that they knew how to find and use an integrating factor to solve a first order differential equation, and many gained either full marks for the question or just lost the final accuracy mark. The only other error of note was losing the negative sign in setting up the integrating factor. This led to a more complicated integral which few solved 'correctly'.

**Q7.**

This question was answered well with most candidates able to show that they knew how to find and use a correct integrating factor to solve a first order differential equation. The majority of candidates then correctly used substitution or inspection to complete the integration. Those candidates who tried to integrate by parts often made errors or did not appreciate that two applications were required. The minority of candidates who correctly expressed  $\sin 2x \sin x$  as the difference of two cosines generally had less difficulty in completing the integration correctly.

**Q8.**

Approximately half the candidates scored full marks for this question. Almost all candidates were able to show that they knew how to find and use an integrating factor to solve a first order differential equation. However, a significant minority did not recognise

the form of the integral of  $\int x^3 (x^4 + 3)^{\frac{3}{2}} dx$  as  $k(x^4 + 3)^{\frac{5}{2}}$  or did not apply a suitable substitution, so made very little further progress.

**Q9.**

Most candidates were able to show that  $\frac{1}{x^2}$  was an integrating factor for the given first-order differential equation and the majority of them went on to correctly find its general solution. A small minority of candidates failed to insert the constant of integration and ended with a general solution which contained no arbitrary constant.

Candidates are advised to check that their general solutions contain as many arbitrary constants as the order of the differential equation being solved. The error ' $y = x^{-2} (\ln x + c)$ '

following a correct previous line ' $\frac{y}{x^2} = \ln x + c$ ' was seen more often than expected.

**Q10.**

Most candidates were able to show that they knew how to find and use an integrating factor to solve a first order differential equation. However, a significant number failed to find the correct integrating factor because they missed the negative sign and used  $e^{\int \tan x \, dx}$ . Candidates should be aware that, unless told otherwise in the question, it is acceptable to leave the solution in a form other than  $y = f(x)$ . Some candidates lost the final accuracy mark because either they had attempted to divide throughout by  $\cos x$  but forgot to divide the  $+ c$  as well before substituting in the boundary condition, or had made the arithmetical

error " $2 = -\frac{1}{2} + c \Rightarrow c = \frac{3}{2}$ ".

**Q11.**

Although many candidates were able to write down the integrating factor in terms of an

integral, a significant minority could not then integrate  $\frac{4x}{x^2 + 1}$  correctly. Those who found the correct simplified integrating factor generally used it appropriately and either solved the resulting integral by a suitable substitution or, more frequently, just multiplied out and integrated  $x^5 + 2x^3 + x$ . Although some candidates failed to insert the constant of integration and so lost the final two marks, this was not a common error.

### Q12.

Most candidates scored full marks for their solution to part (a). Again, part (b) was answered well by the majority of candidates, but it was not uncommon to find solutions which used the wrong integrating factor,  $x$ , or lacked an arbitrary constant. Those

candidates who started part (c) by equating their answer for (b) to  $\frac{dy}{dx}$  and integrating normally scored both marks, although some lost the accuracy mark because their general solution of this second order differential equation did not contain two arbitrary constants.

### Q13.

The vast majority of candidates answered part (a) correctly and also used it to write the left- hand side of the differential equation as an exact derivative. However, finding the

resulting integral,  $\int 3x^2 (x^3 + 1)^{\frac{1}{2}} dx$ , caused problems for a majority of candidates. It was disappointing to see so many failing to spot this integrand as the derivative of

$\frac{2}{3} (x^3 + 1)^{\frac{3}{2}}$  with some resorting to writing  $(x^3 + 1)^{\frac{1}{2}}$  as  $\left(x^{\frac{3}{2}} + 1\right)$ . Subsequent credit, for method, was given to those candidates who introduced a constant of integration and found its value by applying the given boundary conditions.

### Q14.

This question on solving a first-order differential equation by use of an integrating factor was the best-answered question on the paper. Only a few candidates failed to find a correct integrating factor; such candidates failed to realise that the integral of  $\tan x$  is available from the AQA formulae booklet. The most common errors were forgetting to multiply the right-hand side of the given differential equation by  $\sec x$ , or rearranging ' $y \sec x = \tan x + c$ ' incorrectly as ' $y = \tan x \cos x + c$ ' before substituting the given boundary conditions.

### Q15.

This question was a good source of marks for many candidates. The methods in part (a) were well understood and most candidates worked accurately to obtain the correct approximations. In part (b)(i) a high proportion of the candidates obtained the printed answer but some failed to show sufficient stages in their working. In part (b)(ii), weaker candidates could not integrate  $\ln x$ ; such candidates frequently gave the answer as  $1/x$ . Those who inserted the constant of integration were generally able to apply the given boundary condition. Surprisingly, not all candidates who obtained the correct solution for part (b)(ii) were able to give the correct value for  $y$  in part (b)(iii).

### Q16.

The vast majority of candidates started correctly by writing  $= e^{\int \cot x \, dx}$ , and most went on to score the 3 marks. Some candidates, however, seemed to be unaware that  $\int \cot x \, dx$  is given in the formulae booklet and can be quoted directly. In part (b) some candidates had problems integrating  $2\sin x \cos x$ , whilst others used inspection, converting to  $\sin 2x$ , substitution or integration by parts. Those who then divided throughout by  $\sin x$  to get  $y$  explicitly sometimes failed to divide the term '+ c'. In finding the constant of integration it was not uncommon to see  $-\frac{1}{2} \cos \pi$  evaluated incorrectly as  $-\frac{1}{2}$  or even 0.

### Q17.

This was a good source of marks for many candidates. Part (a) was answered well although lengths of solutions varied greatly and a few candidates failed to insert sufficient steps to convince the examiners that the printed answer had been obtained convincingly. In part (b), although the complementary function/particular integral approach was applied correctly by a number of candidates, use of the integrating factor was generally more successful. However some weaker candidates failed to multiply the right-hand side by the integrating factor before proceeding. The start of part (c) begins with 'Hence' without an 'or otherwise' so candidates were expected to use the previous results. Those candidates who tried to solve the differential equation directly generally failed to find the correct particular integral. Many candidates who used the correct method went on to score full marks but some others forgot to insert the constant of integration and ended with the general solution of a second-order differential equation with only one arbitrary constant.

Also, surprisingly,  $\int^x dx$ , was not always found correctly.